AI DATA CENTRES

Power, not chips, is the new bottleneck

Where the real business opportunity sits for IPPs, EPCs and investors

THE GAP

Fast to build.
Slow to power.

18–24 months

to build an AI campus

~128 weeks

for a large power transformer (Wood Mackenzie)

5–10 years

for a grid connection in major hubs

ROOT CAUSE

Why the gap exists

- AI demand arrives in 1–3 years; grids are planned over 5–10
- Transformer and turbine factories were sized for flat 2010s demand
- Interconnection queues are clogged with speculative requests
- Campuses now draw 1–5 GW, the size of a power plant

WORKED EXAMPLE

Firm solar for 1 GW of load

5.2 GW
solar PV

19 GWh
battery storage

~$6 B
total, Masdar/EWEC Abu Dhabi

Rule of thumb: about 5 MW solar + 19 MWh storage per MW of round-the-clock load

OPPORTUNITIES

Where to play

- IPPs: solar + BESS energy parks beside the load
- EPCs: 220/400 kV substations, BESS integration, commissioning
- Batteries in place of diesel backup
- Storage that rides through grid faults and smooths AI load swings
- India (~1.8 GW live, ~3.9 GW pipeline) and the GCC

SOURCES

Data behind this carousel

- IEA, Energy and AI (2025) and 2026 update
- Wood Mackenzie transformer study via pv magazine (2025)
- Masdar/EWEC RTC project: GreentechLead, Renewable Watch (2026)
- ESS News on BESS at data centres (2026)
- Cushman & Wakefield India data centre report (H1 2026)

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